



Traditional Ecological Knowledge of the Arhuaco People on Water Management in the Sierra Nevada de Santa Marta

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Abstract

This article analyzes the role of Traditional Ecological Knowledge (TEK) of the Arhuaco people in water management in the upper Ariguani river basin, in the Sierra Nevada de Santa Marta (Colombia), in a context of accelerated socio-environmental transformation. From an ethnohydrological approach, the study adopts a mixed-methods design that integrates community surveys, participatory techniques, spatial analysis of land use change (2000–2023), and dynamic socio-hydrological modeling. A TEK Index was constructed to evaluate three dimensions of water management: cultural management, indigenous norms, and government norms. The results show that traditional water knowledge is mainly based on ritual practices and indigenous norms guided by indigenous spiritual authorities, while coordination with state institutional frameworks is limited. Spatial analysis shows a loss of forest cover and expansion of agricultural uses around the sacred sites that sustain TEK. Modeling indicates that, although indigenous norms slow the deterioration of the system, only co-governance scenarios can cushion pressures and avoid trajectories of collapse. Taken together, the findings show that the effectiveness of TEK depends on its articulation with multiscale governance arrangements.

Keywords Traditional ecological knowledge · Water governance · Arhuacos · Hydrosocial dynamics · Sierra Nevada de Santa Marta · Colombia

Introduction

Traditional Ecological Knowledge (TEK) constitutes a dynamic system of knowledge, practices, and values that emerges from long-term interactions between Indigenous peoples and their environments (Ngonyani et al., 2024). Rather than a static body of empirical knowledge, TEK expresses relational understandings of territory, water, and non-human beings through symbolic, spiritual, and normative dimensions that guide the use, management, and protection of forests, water, and soils (Berkes, 2018; Tengö et al., 2014; Toledo & Barrera-Bassols, 2008). Originally developed within ecological anthropology, ethnobiology, and human ecology, the concept has expanded into political ecology, environmental governance, and socioecological systems research, although some authors argue that it does not fully capture the spiritual, territorial, and political dimensions of Indigenous knowledge systems, which continue to evolve through interactions with state institutions, markets, and territorial conflicts (De La Cadena, 2010; Ulloa, 2011).

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Within these knowledge systems, water occupies a central ontological and territorial role. Indigenous peoples conceive rivers and sacred sites as living territorial entities rather than merely utilitarian resources, while ritual practices associated with water reinforce territorial memory, intergenerational knowledge transmission, and collective responsibilities toward sacred landscapes (Boelens et al., 2006; Escobar, 2016; Salmond, 2014; Strang, 2004; Aguilera et al., 2023; Valderrama & Escalante, 1988; Gelles, 2000). Growing concerns over water scarcity, climate variability, and ecosystem degradation have renewed scientific and policy interest in these knowledge systems. The UNESCO World Water Development Report (2023) estimates that between two and three billion people experience water shortages for at least one month each year and that global water demand could increase by more than 30% by 2050. Similarly, the IPBES assessment (2024) identifies land-use change, population growth, and ecosystem degradation as key drivers transforming socio-environmental water systems, particularly in rural and Indigenous territories.

This increasing recognition reflects not only international policy agendas but also decades of Indigenous territorial struggles and demands for autonomy across Latin America that have challenged technocratic approaches to water governance by defending territorial rights, sacred landscapes, and alternative understandings of water and nature (Boelens et al., 2006; Ulloa, 2011; Lizarralde et al., 1987; Gelles, 2000; Strang, 2004). Nevertheless, centralized models of water governance have frequently marginalized locally rooted knowledge systems and territorial management practices (Jackson, 2018). Rather than representing a simple divide between Indigenous and state governance, water management typically involves negotiated interactions among Indigenous organizations, farmers, local communities, municipalities, environmental authorities, and other water users operating across multiple institutional and territorial scales. Understanding these interactions remains a key challenge for contemporary water governance research.

Contemporary debates on the relationship between TEK and water governance encompass multiple perspectives on how water, society, and territory interact. Institutional approaches such as Integrated Water Resources Management (IWRM) emphasize multilevel coordination and planning but generally treat water as a technical-administrative resource, limiting the incorporation of Indigenous cultural, spiritual, and normative values (Folke et al., 2005; Hoverman & Ayre, 2012; Olsson et al., 2004). In contrast, hydrosocial, political ecology, and anthropological perspectives highlight water as a socio-natural and relational entity shaped by power relations, territorial disputes, historical processes, and Indigenous ontologies (Boelens et al., 2016; Budds, 2009; De La Cadena, 2010; Salmond, 2014;

Strang, 2004). Likewise, socio-hydrology has advanced the understanding of the coevolution of hydrological and social systems under environmental change, although Indigenous knowledge has generally remained peripheral within these models (Budiman et al., 2023; Kumar & Saizen, 2023).

Ethnohydrology has recently emerged as a complementary framework that integrates these perspectives by recognizing water as a relational entity with ecological, cultural, spiritual, and normative dimensions while linking Traditional Ecological Knowledge with governance processes, biophysical dynamics, and analytical modeling (Coletta et al., 2024; Datta et al., 2025; Diver et al., 2022). This approach enables the analysis of interactions among local management practices, governance arrangements, and environmental change, including the effects of land-use transformation, demographic dynamics, and institutional marginalization, while highlighting the potential of intercultural governance to strengthen environmental management in socioecologically transformed territories (Coletta et al., 2024; Curran, 2019).

From this perspective, ethnohydrology provides an appropriate framework for understanding territories where Indigenous ontologies of water coexist with rapid socio-environmental transformations and multiple governance systems. The Sierra Nevada de Santa Marta (Niwiumuke), Colombia, represents one such case, as it supports regional water supply, biodiversity conservation, Indigenous governance, and diverse actors with competing interests over land and water management.

The Sierra Nevada de Santa Marta (Niwiumuke) currently faces increasing deforestation, agricultural expansion, demographic change, and territorial disputes that affect watershed integrity and hydrological processes sustaining both Indigenous and rural populations (Molina-Parra et al., 2025). These transformations have intensified interactions among Indigenous communities, farmers, municipalities, and environmental authorities while increasing the overlap between Indigenous territorial governance and state environmental planning instruments. Despite these pressures, ritual water management systems based on sacred sites, the Law of Origin, and Indigenous territorial governance remain central to water management and coexist with agricultural systems, state regulations, and broader environmental governance processes (Guáqueta-Solórzano, 2022).

Against this background, this study examines the relationship between Traditional Ecological Knowledge and water management in the upper Ariguaní River basin, a sacred territory of the Arhuaco people. Adopting an ethnohydrological perspective, the analysis integrates ritual practices, sacred sites, collective norms, land-use change, demographic dynamics, and the interaction between Indigenous and governmental governance systems to understand

how socio-environmental transformations reshape water governance (Acharibasam et al. 2024a, 2024b; Berkes 2018; Datta et al. 2025; Diver et al. 2022; Genova and Wei 2023; Lillo-Saavedra et al. 2024). Specifically, the study addresses three questions: (i) What is the current state of TEK and which ritual and territorial practices shape water management in the upper Ariguani River basin? (ii) How have land use and demographic dynamics transformed the sacred sites associated with TEK? and (iii) How do Indigenous and state governance systems interact under increasing socio-environmental pressures?

Materials and Methods

Study Area

The research was conducted in the four indigenous settlements of the upper Ariguani River basin: Mamarwa, San Pablo, Nabaku, and Guanani. This river originates at an altitude of approximately 2,779 m above sea level, flowing

through the departments of Cesar and Magdalena and forming part of a basin with an area of 4,553.8 km². In its upper course, it reaches a length of 27 km and has a dendritic drainage pattern determined by its passage over metamorphic and sedimentary formations (Fig. 1). The upper Ariguani River basin constitutes not only a hydrological unit but also a culturally, politically, and legally significant Indigenous territory where state institutions, Indigenous governance systems, and rural livelihoods coexist and interact.

The upper basin crosses the expansion area of the Arhuaco Indigenous reserve (*resguardo*) and part of the rural area of Pueblo Bello municipality, Cesar. Indigenous reserves are legally recognized collective territories through which Indigenous peoples exercise territorial autonomy and self-government. The Arhuaco reserve covers 201,263 ha, while the 2,000-ha expansion area comprises four landholdings acquired in 2020 as part of a territorial recovery strategy. Since then, approximately 100 Indigenous families have progressively settled in the area, which had previously been occupied by farmers and already showed deforestation and water degradation. Today, Indigenous and farming

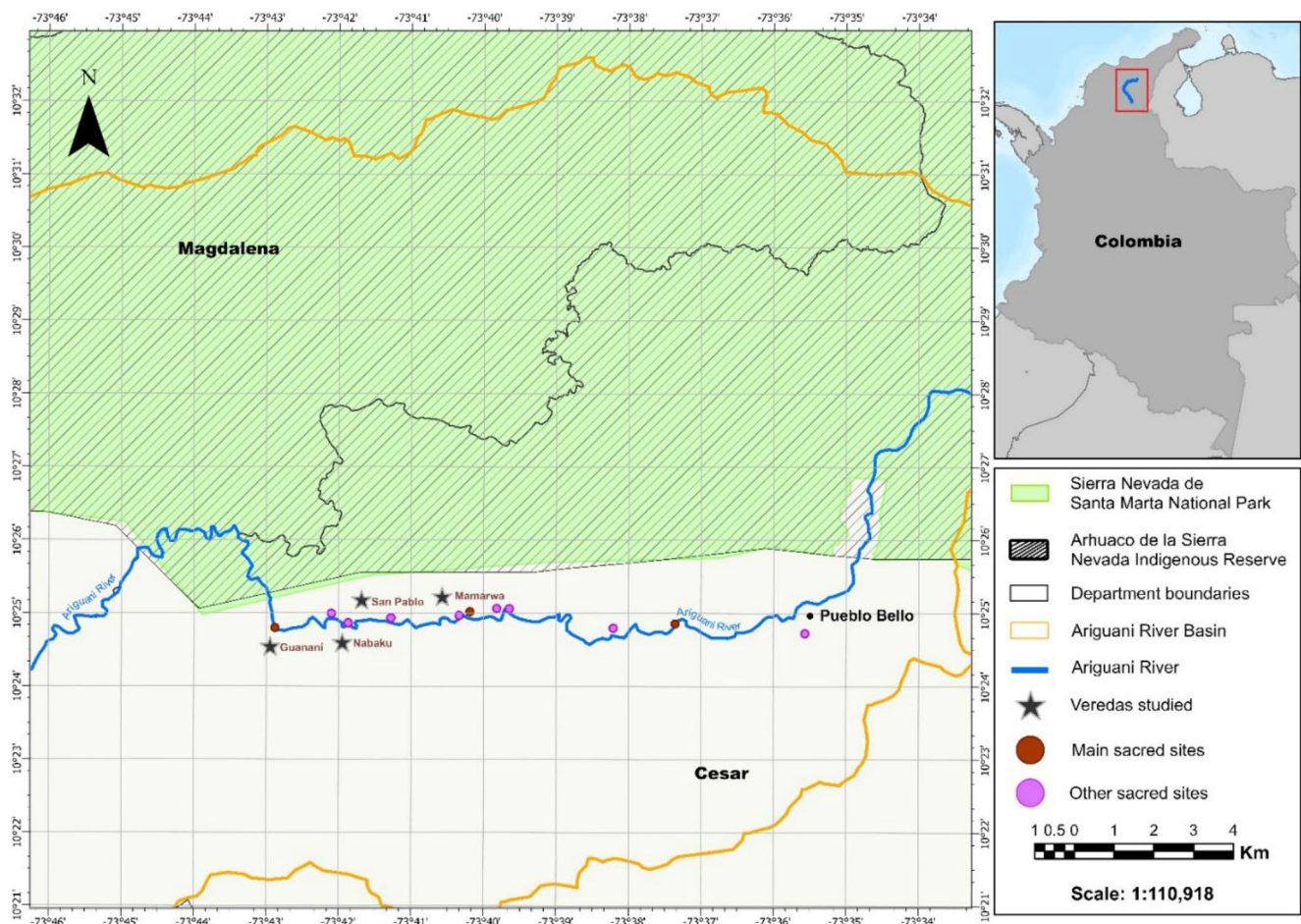


Fig. 1 Study area of the Ariguani River. The villages of San Pablo, Mamarwa, Nabaku, and Guanani, sampling sites for the study, and the location of sacred sites identified by the community are shown

families coexist within the basin, creating a governance context where different land-use systems, environmental values, and institutional arrangements overlap.

Indigenous livelihoods are based on subsistence agriculture and small-scale livestock, whereas farmers cultivate coffee (*Coffea arabica*), cocoa (*Theobroma cacao*), and avocado (*Persea americana*) on smallholdings of less than 2 ha, contributing to the fragmentation of tropical dry forest remnants (Alcaldía Municipal de Pueblo Bello, 2020; Pertuz Peralta & Pérez Orozco, 2017). Occasional small-scale sand extraction also generates localized pressures on riverbanks and water dynamics. The Arhuaco population represents approximately 40% of the inhabitants in the study area, reflecting the coexistence of Indigenous and non-Indigenous communities within the same watershed.

Watershed management falls under the River Basin Management and Planning Plan (POMCA), the main state instrument for integrated watershed planning in Colombia, promoted by the environmental authorities of Cesar and Magdalena to coordinate environmental planning, water governance, and territorial management. However, the basin is highly transformed, with only 15% of its area maintaining continuous forest cover, while 85% has been altered by agricultural expansion, land-use change, and dispersed infrastructure (Viloria De La Hoz, 2019). Moreover, the POMCA has not yet been updated or implemented, creating governance gaps between state planning and Indigenous territorial management, particularly regarding water conservation and land-use regulation.

Arhuaco social organization is based on extended families grouped into communities governed through an integrated system of spiritual and political authorities. The Mamus, as the highest spiritual authorities, safeguard the Law of Origin (*Ley Sein zare'zanu*), an ancestral normative system that regulates the relationship between people, nature, and the sacred world. Through ritual practices such as *pagamentos* and *saneamientos*, they guide collective decisions and maintain territorial and ecological balance (González et al., 2021; MAPP-OEA, 2008).

Cabildos constitute the institutional interface between Indigenous governance and the Colombian state through the Special Indigenous Jurisdiction, enabling Indigenous communities to exercise territorial governance according to their own normative systems within their ancestral territories. Although the 1991 Constitution formally recognized Indigenous territorial rights and autonomy, tensions between Indigenous authorities and state institutions over environmental and territorial governance persist. Consequently, water governance in the Ariguani basin reflects an intercultural co-governance process shaped by overlapping normative systems and unequal power relations.

The Law of Origin (*Ley Sein zare'zanu*) is the guiding spiritual and normative framework of the Arhuaco people, articulating a pedagogy of reciprocity in which nature is understood as a network of interdependent beings rather than a resource (Guáqueta-Solórzano, 2022). Transmitted by the Mamus, who are trained from childhood in sacred sites, it establishes the collective responsibility to maintain the balance between people, nature, and the spiritual world. Consequently, water governance is conceived not merely as an environmental or technical issue but as a territorial and spiritual responsibility.

Sacred sites (*Tana*) constitute the core of Arhuaco water governance, as they are regarded as living spaces inhabited by the mothers or spirits of water that sustain territorial balance. The Mamus interpret changes occurring at these sites to guide ritual practices such as *pagamentos* and *saneamientos*, making them fundamental spaces for territorial regulation.

Methodology

Qualitative and quantitative data were collected between July 6 and 12, 2025, in four Arhuaco settlements (San Pablo, Mamarwa, Nabaku, and Guanani) located along the upper Ariguani River. Prior authorization was obtained from the Tayrona Indigenous Confederation, the Mamu of Mamarwa, and the traditional authorities of the territory (June 26, 2025). A total of 48 surveys (16 men and 32 women) were administered to Arhuaco community members selected by traditional authorities. The structured questionnaires, conducted in Spanish under informed consent, assessed three TEK dimensions adapted from Acharibasam et al. 2024a, 2024b; Diver et al. (2022): cultural management, Indigenous norms, and government norms. Because participants were identified by Indigenous authorities, the sample was purposive rather than statistically representative.

To complement the surveys, five semi-structured interviews with Indigenous leaders, two focus groups with Mamus, and two participatory workshops involving 32 participants (10 women and 22 men) were conducted. Workshops applied Participatory Geographic Information Systems (PGIS) to identify sacred sites, river tributaries, human settlements, economic activities, and environmental threats affecting the Ariguani River (Dunn, 2007).

Analysis of the Information

To characterize TEK, we constructed an index that we call the Traditional Ecological Knowledge Index (TEK Index). As can be seen in Table 1, we used 16 variables that were tracked in the survey and normalized using a Min-Max

Table 1 Dimensions and variables for the analysis of TEK related to water management

Dimension	Variables
Cultural management	Water rituals; conception of water as a living being; traditional practices of use; existence of sacred places; transmission of knowledge; threats to knowledge; threats to sacred places; perceptions of water quality; traditional authorities who apply knowledge
Indigenous norms	Community rules on access and use; compliance with own rules; community sanctions; participation of traditional authorities
Government norms	Legal recognition of indigenous norms; inter-institutional coordination to conserve and manage the river; implementation of water policies; regulatory conflicts

rescaling scheme to have variables on a similar measurement scale between 0 and 1 (0 being the worst range and 1 being the best value of a variable). The range of the index is explained from 0 to 3 (total dimensions studied).

The TEK index was calculated for the Arhuaco community on the following scale:

$$TEK = \sum_i^1 D_i$$

Where TEK is the result of the sum of the averages of D_i (average of the total variables analyzed in the Dimensions). The D_i is obtained as follows:

$$D_i = \frac{1}{n} \sum_k^1 x_{ik}$$

Where D_i is the result of the average of the total sum of the normalized variables (i) of each dimension.

Although closely related, cultural management refers to ritual and everyday cultural practices associated with water, whereas indigenous norms refer to community rules and regulatory mechanisms governing access to water and sacred sites. Participants recognized both as interconnected dimensions of water governance.

Boxplots were plotted with the index results to visualize the differences in medians, outliers, and data dispersion, using RStudio version 2025.09.0+387 (Posit team, 2025).

To analyze the dynamics of socio-environmental change associated with land use change and population growth, the scale of the four settlements was expanded and the upper basin was taken as a reference to show a broader change. This decision was made based on the recommendation of Indigenous leaders who considered that the river is a structure that goes beyond their settlements. To this end, a landscape analysis was carried out using the MapBiomass database (version 2023). The annual series 2000–2023 were

used for the departments of Cesar and Magdalena, cropped to the study area using ArcGIS Pro version 3.4.0 (ESRI, 2024) with a buffer of 300 m on each side of the river. Spatial processing included the reclassification of relevant land cover types, the construction of transition matrices between reference years, and the calculation of areas and percentages of change by class in RStudio (version 2025.05.1+513).

The choice of a 300-meter buffer is based on ecological evidence showing that narrow riparian strips (<50 m) are insufficient to preserve complex ecological processes and associated biodiversity, while widths of around 200–380 m allow for the integration of hydrological, microclimatic, and habitat connectivity functions (De Oliveira Ramos & Anjos, 2014; Sweeney & Newbold, 2014). This area was corroborated through consultation with the community, which pointed out the need to maintain a prudent distance from the riverbed due to the presence of sacred sites.

For the demographic analysis, changes in the number of inhabitants in the upper basin were analyzed using available data between 2005 and 2023 for the farmer population (DANE, 2016) and between 2012 and 2025 for the Indigenous population. The latter period corresponds to the timeframe in which Indigenous leaders identified major landscape transformations and the arrival of Indigenous families in the expansion area. Due to the absence of official census data, Indigenous population estimates were based on information provided by community leaders.

Finally, a dynamic socio-hydrological model was developed in NetLogo version 6.3.0 (Wilensky, 1999) with the aim of exploring how variations in TEK, expressed through cultural management, Indigenous norms, and government norms, shape different water governance arrangements in the sacred sites of the upper Ariguaní River basin. The model was structured around three contrasting scenarios: (i) a baseline scenario, without cultural management or community or state regulation; (ii) an ethnohydrological scenario, in which cultural management and indigenous norms predominate; and (iii) a co-governance scenario, which articulates indigenous and state norms through cooperation mechanisms. This configuration allowed for a comparative analysis of how different normative expressions of TEK influence the dynamics of the system.

The model integrates three types of actors: the Arhuaco people, who regulate water use through traditional norms, rituals, and community sanctions; farmers, associated with pressures derived from agricultural expansion, deforestation, and water abstraction; and the state, which is associated with environmental authorities that intervene through environmental policies and regulations. These actor categories do not represent homogeneous or isolated groups, but rather territorial actors identified by the Arhuaco community as directly involved in water management and land-use

dynamics within the basin. Farmers are primarily associated with agricultural uses oriented toward coffee cultivation and other productive activities, while Indigenous communities emphasize spiritual and territorial forms of water regulation linked to sacred sites. Although tensions occasionally emerge regarding the deterioration of sacred places and water sources, interactions between Indigenous and peasant populations are also shaped by everyday coexistence and ongoing social exchange within the territory. Based on these interactions, the temporal evolution of the state of sacred sites was simulated, represented as a continuous variable between 0 (degraded state) and 1 (conserved state), depending on the balance between normative benefits and ecological pressures.

Regulatory benefits derive from compliance with Indigenous norms, state norms, and cooperation between the two systems, while ecological pressures represent the effects of intensive land use, changes in vegetation cover, and population growth. The model incorporates social feedback mechanisms that capture, on the one hand, the reinforcement of Indigenous norms in the face of degradation of sacred sites and, on the other, the modulating effect of cooperation on the effectiveness of state regulation. Environmental, anthropogenic, and demographic disturbances were also included to represent processes of deforestation, agricultural expansion, and population growth.

The model parameters were defined based on a literature review and field observations, seeking to reproduce the asymmetry observed in mountain systems, where

degradation processes tend to be faster than recovery processes. A sensitivity analysis ($\pm 20\%$) was performed to assess the stability of the system in the face of parametric uncertainty, identifying greater sensitivity to the strength of Indigenous norms and farmer pressure. Overall, the modeling enabled the exploration of how different regulatory arrangements influence the persistence or deterioration of sacred sites under scenarios of increasing socio-environmental pressure (see supplementary material A for the formal model equations).

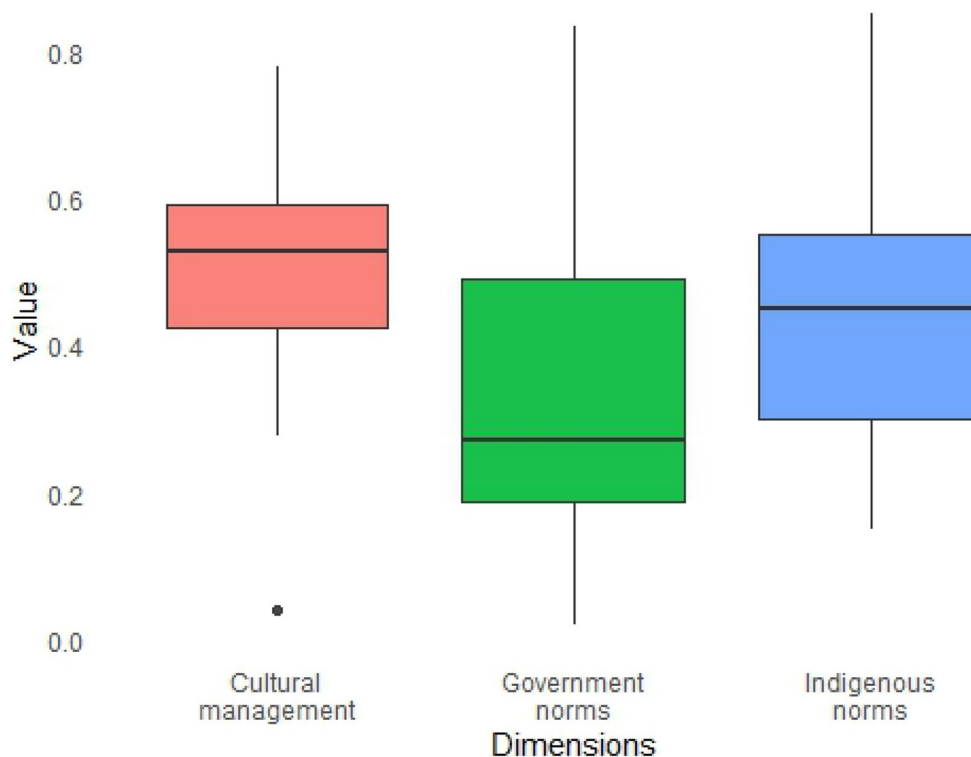
Although this study incorporates a modeling component, the empirical information used was supported by approximately six years of previous research experience in the territory, which facilitated prior knowledge of the study area, sustained interaction with Arhuaco communities, and familiarity with local environmental and governance dynamics.

Results

Traditional Ecological Knowledge of Water

The aggregate TEK Index obtained a total value of 1.28, reflecting a low level of knowledge in relation to water management (Fig. 2). This result reflects community perceptions regarding current limitations in institutional recognition, pressures affecting Indigenous norms, and external influences on cultural practices that constrain water management based primarily on TEK.

Fig. 2 Distribution of TEK Index values by dimension



The disaggregated analysis of the index shows clear differences between the three dimensions evaluated. The government norms dimension had the lowest value (0.34), followed by the Indigenous norms dimension (0.45), while the cultural management dimension achieved the highest score (0.50). However, even the latter is far from the maximum threshold of the index, indicating that none of the dimensions is fully consolidated.

Cultural Water Management: Centrality of the Law of Origin

The cultural management dimension obtained the highest score within the TEK Index (0.50), indicating that cultural knowledge continues to be the strongest and most operational component of TEK. However, its intermediate value suggests that cultural management currently coexists with additional social, environmental, and institutional pressures affecting water governance in the basin.

Thus, the community affirms that its relationship with water is governed by the Law of Origin, understood as the normative and spiritual framework that orders life, territory, and the relationship between human beings and nature. Under this Law, rivers, springs, and streams are linked to sacred places that sustain spiritual and territorial balance. These spaces not only regulate the flow of water but also ensure harmony between the different elements of the territory. From the Arhuaco perspective, balance and harmony are not understood as static environmental conditions, but as the continuous maintenance of reciprocal relationships between water, sacred sites, human actions, and ancestral spiritual beings through ritual and territorial practices.

Among these practices, *saneamiento* and *pagamento*, guided by the *Mamus*, stand out. *Saneamiento* corresponds to spiritual cleansing actions that are performed when a sacred site is disturbed or the balance of the territory is affected, for example, by contaminating a spring, intervening at a point on the river, or entering a sacred space without authorization. Subsequently, *pagamento* restores spiritual order through symbolic offerings, such as cotton, leaves, stones, water, and elements of the territory, deposited at specific sacred points.

The community distinguishes between different types of *pagamentos* depending on the nature of the imbalance. *Anuwe Duna* is related to positive thoughts offered to sacred sites, while *Anuwe Gansigna* corresponds to negative thoughts that must also be delivered to balance energies. The *Tikrike pagamento* is performed when it is necessary to repair damage caused to the territory, such as fires, damage to mountains, or degradation of areas near springs.

Performing these rituals requires rigorous spiritual preparation called *Marneik*, which involves dietary abstinence

and a specific physical and spiritual disposition. Compliance with these rules is verified by the *Mamu*, who may establish additional restrictions to ensure that balance is effectively restored.

Likewise, from the Arhuaco perspective, the Sierra Nevada in its entirety constitutes a sacred site, whose Law of Origin permeates and orders the entire territory, articulating the material and spiritual worlds. Within this framework, *pagamentos* and *saneamientos* are understood as spiritual management practices aimed at sustaining territorial balance, so that water management is not limited to protecting catchment points, but also maintains a network of relationships and obligations. Material water management practices include a rule whereby 70% of the territory is conserved and 30% can be used for subsistence crops.

Despite the centrality of the Law of Origin and the diversity of cultural practices associated with water care, the community recognizes that cultural management faces significant limitations. The intergenerational transmission of knowledge is not fully assured, as many young people spend less time in the territory. Added to this are external pressures on the water system, such as decreased flow and pollution generated by the farmers who inhabit the territory, which exceed the capacity of natural-cultural regulation and limit the capacity of TEK-based practices to respond independently to increasing socio-environmental pressures.

The community also recognizes that water management practices are increasingly shaped by processes of social and cultural interaction between indigenous and peasant families, generating new understandings of water associated with commercial agriculture, domestic use, and changing territorial dynamics. In this sense, contemporary water management in the territory is not based exclusively on traditional knowledge, but rather on the coexistence of different forms of knowledge and environmental practices. However, participants emphasized that limited institutional accompaniment and weak intercultural coordination have reduced opportunities for more collaborative forms of water governance between indigenous and state actors.

From the community's perspective, successful water management does not imply the absence of water scarcity, but rather the maintenance of territorial and spiritual balance between water sources, sacred sites, and everyday human activities. Participants did not express generalized dissatisfaction regarding access to water for domestic needs, although they emphasized that prolonged drought periods, increasing population growth, and agricultural expansion have increased pressure on local water sources. Community members particularly identified pollution associated

with agricultural activities and the progressive deterioration of sacred sites as the main current threats to water management. Participants also noted that stronger municipal environmental control and greater coordination with state authorities could contribute to reducing environmental pressures affecting the basin.

Government Norms: Lack of Recognition and Institutional Appropriation

The government norms dimension was the weakest in the index, reflecting a limited perception of the presence and appropriation of state regulatory frameworks in the territory. The community claims not to recognize the existence of formal regulations governing water management, even though it points out that a POMCA is being developed for the watershed to which the territory belongs.

From the community's perspective, the POMCA is not seen as an operational or visible instrument in the territory. The community does not identify any concrete actions associated with this process, nor any mechanisms for coordination with its own forms of water regulation. In this sense, institutional water planning is perceived as an external process, distant and disconnected from everyday life in the territory, with no direct impact on the protection of water flow, the conservation of water sources, or the resolution of environmental conflicts between indigenous and farms communities that arise from differing values regarding water.

This absence should also be interpreted as an intercultural disconnect; institutional frameworks are not perceived as effective regulation because they do not translate into visible actions within the territory and because any external intervention must first be received and harmonized with the Law of Origin. In addition, it must be recognized before sacred spaces that are considered regulators of the territory, in order to prevent unwanted effects and ensure that projects meet their objectives.

The community points out that, in the face of problems such as the decrease in river flow, pollution from agricultural and livestock activities, erosion, the expansion of livestock farming, and sand extraction in the streams, there is no institutional support or inter-institutional participation. In the absence of such support, decisions are made internally through community meetings aimed at analyzing the state of water resources and the factors that affect them. However, these responses do not refer to the implementation of water policies, municipal programs, or actions derived from the POMCA, reinforcing the perception of a disconnect between state water governance and indigenous territory.

Indigenous Norms: Internal Consistency and Restrictions in the Face of External Impacts

The dimension of indigenous norms achieved an intermediate score (0.45), reflecting the existence of clear and coherent internal regulations, but with significant limitations on their application in the face of external actors and dynamics.

The community affirms that it has its own rules for water access and management, as well as sanctions for those who violate these rules. Among the most widely recognized rules are the obligation to keep springs clean, the prohibition of diverting or manipulating water without authorization, the restriction of access to certain sacred sites, the prohibition of cutting down trees near water sources, and the mandatory performance of *saneamientos* or *pagamentos* when changes in water flow, water quality, or the balance of the territory are perceived.

Community sanctions are not conceived as punitive punishments, but as restorative mechanisms aimed at reestablishing spiritual balance. When a person causes damage to a tributary or a sacred place, they must perform a *saneamiento*, followed by a *pagamento* guided by the *Mamũ*. In addition, the person must serve a period of reflection, the length of which depends on the seriousness of the offense, under the supervision of the *semanero* or police and imprisonment in the area known in the indigenous language as *Animakũ*.

These rules are based on the Law of Origin, which the community recognizes as the highest principle governing its relationship with water. However, the community points out that the lack of coordination with state entities limits its ability to enforce these rules in the face of impacts from outside the territory, such as the implementation of mining projects and the progressive occupation of the territory by farmers, which represent a critical threat.

Participants indicated that state authorities formally recognize indigenous communities as territorial authorities with decision-making rights over their territories. However, respondents also emphasized that, in practice, state institutions rarely consult communities regarding watershed management or conduct joint monitoring and planning processes. According to participants, this limited interaction has reduced opportunities for coordinated actions between Indigenous and governmental governance systems.

The community also associates these external influences with increasing social and cultural interaction between indigenous and peasant families, which has gradually transformed traditional water management practices. According to participants, younger generations increasingly adopt agricultural practices oriented toward commercial crops

such as coffee, where water is primarily understood as a productive resource rather than as a spiritual and territorial entity requiring ritual care practices such as *pagamentos* and *saneamientos*.

Traditional Water Management Practices

The research identified two types of spiritual water management practices: *pagamento* and *saneamiento*, and material management practices referred to as cultural restrictions.

Water management is not expressed as an isolated technical action, but as a set of differentiated practices that are activated according to the type of risk identified and the magnitude of the alteration of the territorial balance. A central element of this system is the existence of sacred sites, understood as specific areas of the river and its immediate surroundings, configured by sacred stones and forests, where the spiritual and environmental balance of the territory is concentrated. The research identified 11 key sacred sites near the river, recognized by the community as strategic points for offerings, several of them linked to water (see supplementary material B for photos).

Saneamiento and *pagamento* are the main ritual practices associated with water management and are carried out under the guidance of the *Mamus*, who act as spiritual authorities. Both practices are closely linked, but fulfill different functions within the management system. *Saneamiento* operates as an immediate and corrective response, activated in the face of specific incidents such as localized contamination, unauthorized entry into a sensitive area, or improper handling of a tributary. Its function is to contain the damage and prevent the imbalance from worsening. From the community's perspective, *saneamiento* is seen as a human intervention that emerges from a direct mandate from *Zaku Murun* (Invisible or Intangible Mother).

The *pagamento*, on the other hand, is activated when the disturbance of the balance is more profound or persistent, or when *saneamiento* is insufficient. This practice is restorative in nature, aimed at reordering the territorial balance as a whole and ensuring the continuity of water flow. Both mechanisms are carried out in sacred places and are part of a management sequence that allows for a gradual response to different levels of environmental risk.

However, in terms of sequence, when the event is interpreted as a one-off occurrence, *saneamiento* may be sufficient to stop the spread of that imbalance. However, if the impact is perceived as structural (persistent degradation of the environment), the response tends to escalate to more far-reaching practices, such as *pagamento*, which restore territorial and spiritual order.

On the other hand, cultural restrictions correspond to preventive practices aimed at preventing the risk

from materializing. These material norms are enforced by the *cabildo*, as the authority that governs the settlements, and supervised by the *semanero*, who performs community control functions. The *cabildo* establishes the sanctions and periods of confinement or community imprisonment, defined according to the severity of the damage caused to the water or the territory, while the *semanero* ensures compliance. Restrictions include regulating access to sensitive areas, prohibiting the felling of trees near water sources, and controlling water use, reducing the likelihood of damage and upholding community agreements.

A summary of the traditional spiritual and material management practices associated with the environmental problems perceived by the community can be seen in Table 2.

The water management system under material rules is activated when preventive practices and rituals are not sufficient to prevent or correct damage. These measures, defined by the *cabildo* according to the severity of the damage to water, forests, or sacred sites, include the payment of a sum of money, periods of confinement in the traditional prison for between one and six days, without food and aimed at reflection, as well as public sanctions involving cleaning, maintenance of the territory, and demanding community work.

Beyond ritual practices, water management is embedded in everyday life. Arhuaco families use river and spring water for cooking, hygiene, washing, and irrigating subsistence crops. Water is manually collected from nearby rivers using buckets or traditional containers, as communities lack water distribution and sanitation systems. These daily practices are also regulated through cultural norms and the territorial balance established by the Law of Origin.

Dynamics of Socio-environmental Change: Land Use Change and Population

The Arhuaco people perceive that the main social and environmental changes affecting the Ariguani River basin are associated, on the one hand, with increased deforestation and, on the other, with population growth dynamics in the territory. These dynamics developed before the community arrived in this area, given that these lands correspond to areas of expansion of the indigenous reserve, acquired in 2020. From this perspective, the transformations of the river and the sacred spaces that sustain it can be explained by a greater opening up of the landscape to productive uses and by the increase in the farmer population settled in the basin, which has intensified pressure on water, forests, and sacred sites. This perception is the starting point for analyzing the ecological and social transformations recorded in the upper Ariguani River basin between 2000 and 2023.

Table 2 Traditional water management practices and associated environmental problems

Environmental problems	Type of practice	Responsible authority	Area of application	Main function
Local-ized river pollution	Saneamiento	Mamus	Sacred sites associated with the river	Immediate containment of damage
Fires and deforestation	Pagamento	Mamus	Sacred areas affected	Repair and compensation for damage
Unauthorized access to sensitive areas	Delimitation of areas of non-use	Cabildos and semaneros	Sacred sites and birthplaces	Prevention of damage and generation of lessons to avoid others replicating the problem
Logging near water sources	Imprisonment or payment of fines	Cabildos and semaneros	Sacred forests and water patrols	Preventive protection of water flow
Non-compliance with community regulations	Saneamiento and pagamento	Mamus/cabildos/semaneros	Defined ritual spaces	Corrective measures and individual penalties in the form of fines or imprisonment
Increase in the number of farmers	Spaces for dialogue to avoid conflict	Cabildos and semaneros	Traditional territory	Preventing an increase in settlers and invasions

Between 2000 and 2023, the landscape of the basin underwent substantial changes in its land cover structure (Fig. 3).

The forest was reduced from 524.7 ha to 317.9 ha, representing a total loss of 206.8 ha. At the same time, the mosaic of agriculture and pastures expanded from 718.8 ha to 918.1 ha, with a net gain of 199.3 ha, consolidating itself as the dominant land cover in the territory. Non-forest natural formations practically disappeared, falling from 18.6 ha to 0.62 ha, while urban infrastructure increased from 1.68 ha to 24.21 ha and areas without vegetation grew from 1.33 ha to 4.17 ha. Taken together, these changes confirm a sustained reduction in natural cover and an expansion of productive uses in the upper Ariguaní River basin.

Spatial analysis of land cover changes in relation to identified sacred sites in the basin shows that landscape transformations are not distributed evenly along the river. Several sacred sites are currently surrounded by mosaics of agriculture and pastures. These results coincide with social mapping, in which participants noted that, approximately 20 years ago, there was a greater presence of forest in areas surrounding the upper Ariguaní River basin, although initial processes of forest extraction and the appearance of livestock and farmer populations were already being identified. In contrast, for the current period, social mapping records the intensification of logging, a greater number of water extraction points, the construction of roads, and an increase in the cultivation of plantains (*Musa × paradisiaca*), coffee (*Coffea arabica*), and cocoa (*Theobroma cacao*), as well as livestock (Fig. 4).

In terms of the demographic dynamics of the farmer population in the upper basin, there has been sustained population growth due to births in recent decades. The total population increased from 16,942 inhabitants in 2005 to 27,007 in 2018, with official projections estimating 30,844 inhabitants by 2023, including more than 21,000 residents. This demographic increase coincides spatially with areas where the expansion of the agricultural mosaic has been documented, particularly in the four settlements studied.

Part of this population growth is associated with the arrival of settlers during a wave of migration that began in the 2000s, linked to regional displacement dynamics resulting from the Colombian armed conflict involving illegal armed groups seeking territorial control for illicit crop production. These processes contributed to the demographic reconfiguration of the territory and the expansion of rural settlements, in many cases related to agricultural activities

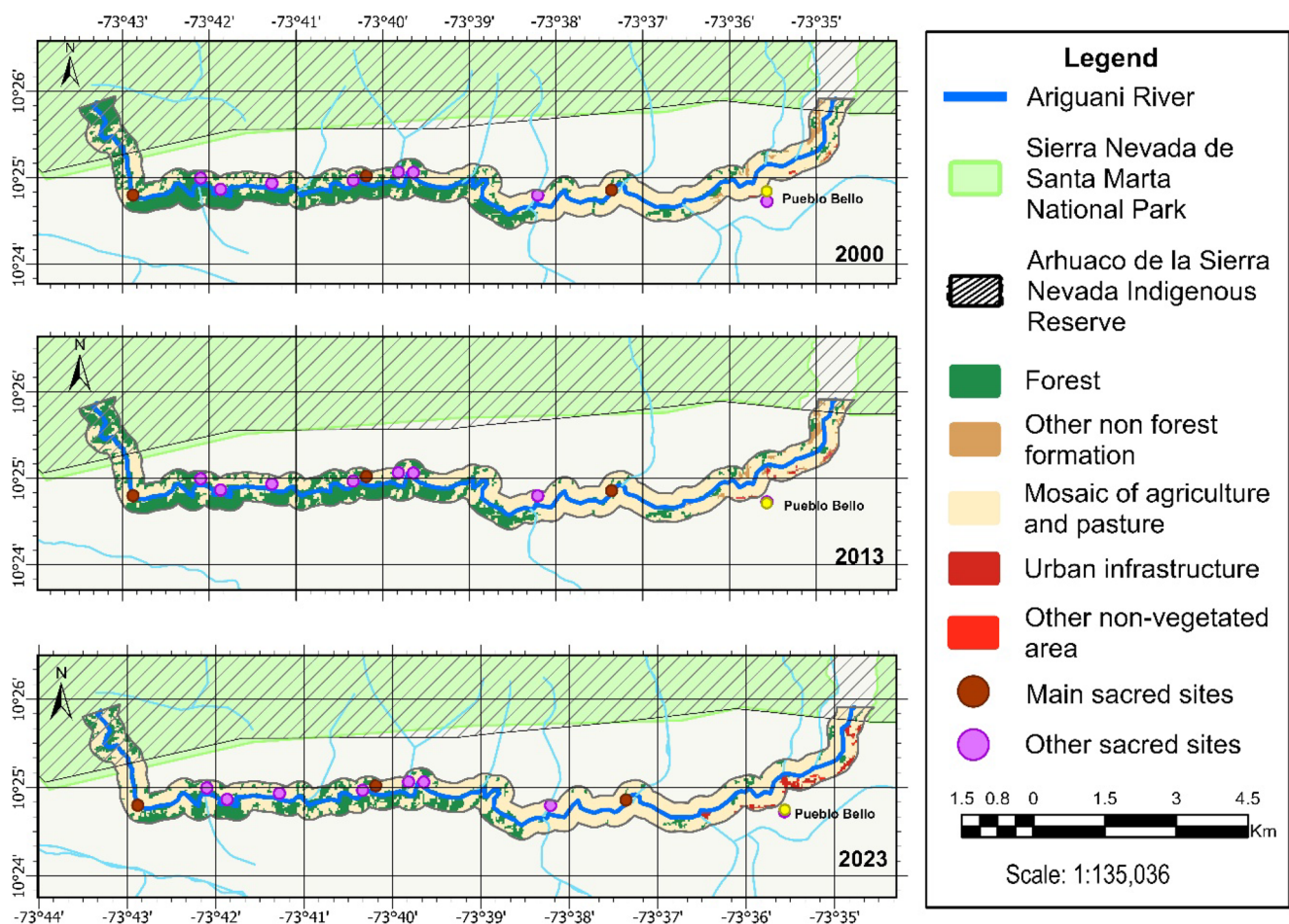


Fig. 3 Changes in land cover in the upper Ariguani River basin (2000, 2013, and 2023)

that coincide temporally with periods of landscape transformation identified in the basin.

As for the indigenous population, in 2010 there were an average of 152 inhabitants, in 2013 there were 237, and in 2023 there were 330. Unlike the farmer population, which shows a higher number of births, this dynamic shows population growth due to the resettlement of families moving down to the areas of expansion.

In summary, the population increase, especially among farmers, has occurred simultaneously with processes of landscape transformation, evidenced by the progressive reduction of forest cover and the increase in intervened areas. Between 2000 and 2023, forest cover decreased while the population showed a growing trend, indicating a temporal convergence between demographic changes and modifications in the landscape structure. This reinforces the perceptions of the respondents, who point out that the progressive environmental deterioration of the watershed and sacred sites occurs as the population increases and that this has led to an erosion of TEK, insofar as it affects the physical spaces where pagamentos and spiritual saneamientos processes take place.

Ethnohydrological Modeling

The three scenarios evaluated allow us to observe how different regulatory arrangements defined based on the dimensions of the TEK Index modify the interaction between farmer pressure (Pf), population growth (Pop), the strength of indigenous norms (Ni), and the state of sacred sites (Ss).

The Fig. 5 shows the joint temporal evolution of these variables for the Base, Ethnohydrological, and Co-governance scenarios over 50 simulation steps, showing how the system responds when the activation of the normative dimensions of the index varies.

In the Negative scenario, characterized by the absence of cultural management, indigenous norms, and government norms (all dimensions of the TEK Index with zero values), Pf increases rapidly and reaches extreme values in the first simulation steps. Under these conditions, Ss declines rapidly, reaching values close to zero before cycle 50. The Ni variable remains at zero throughout the period, reflecting a scenario without active indigenous norms, without cultural management, and without mediation associated with the

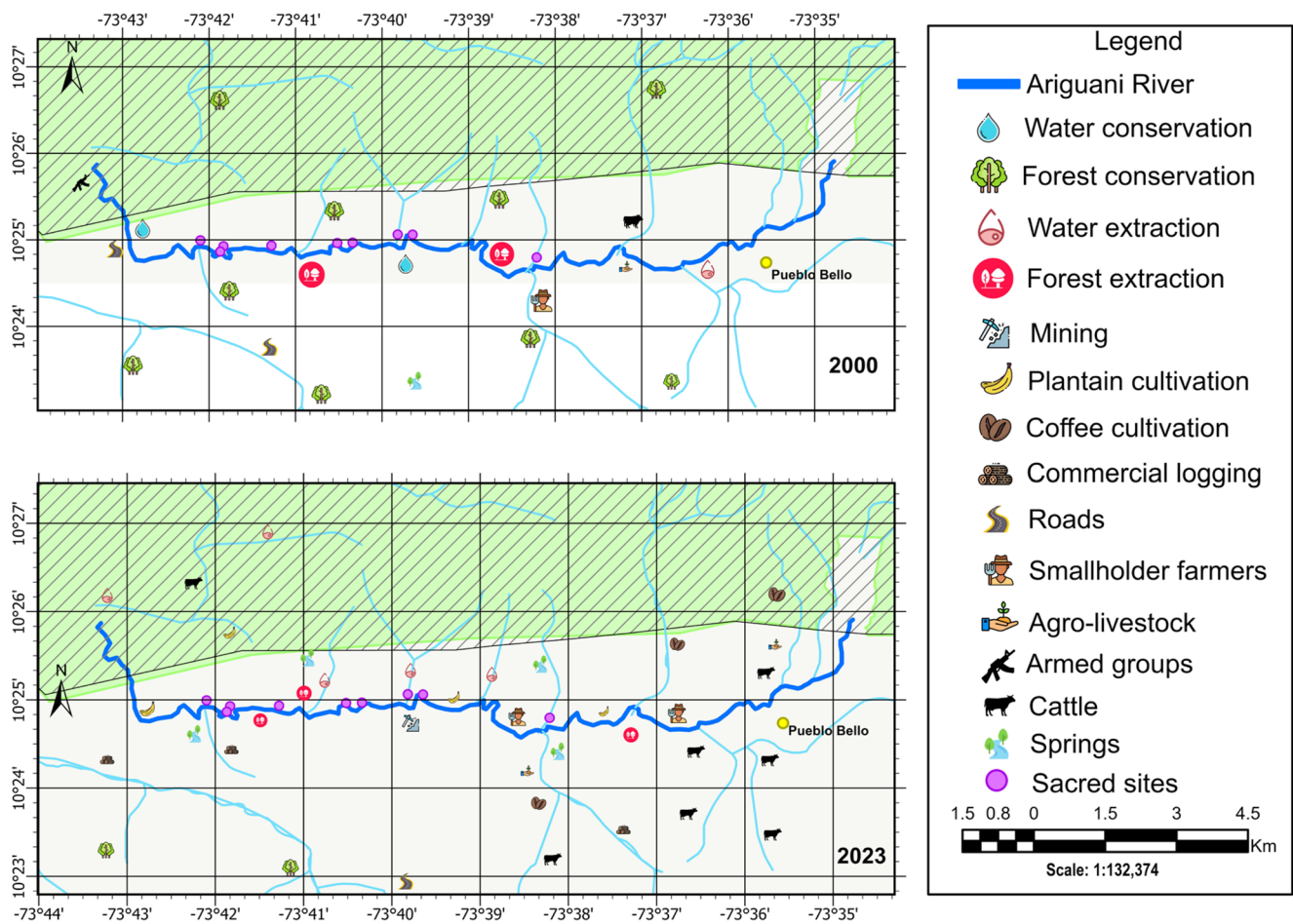


Fig. 4 Social cartography of the Ariguani River basin comparing conditions reported for the years 2000 and 2023. The symbols represent elements identified by Arhuaco participants. The map shows the spa-

tial relationship between sacred sites and the progressive expansion of the agricultural mosaic along the river corridor. Data obtained through participatory mapping, field georeferencing, and GIS analysis

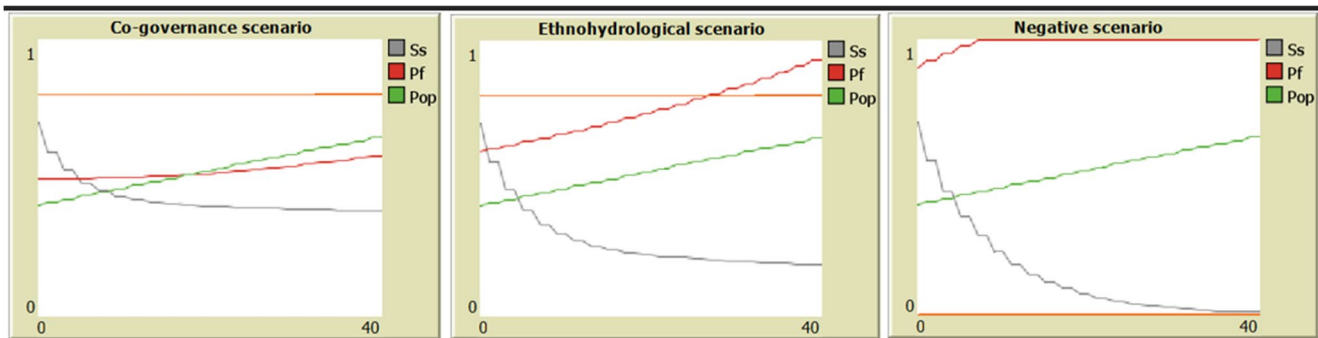


Fig. 5 Temporal evolution of Ss, Pf, and Pop in the Negative, Ethnohydrological, and Co-governance scenarios. Each panel represents 50 simulation steps

Law of Origin. This trajectory reflects a collapse of the cultural and ecological system: high pressure, loss of normative support, and almost total disappearance of sacred sites. This behavior contrasts directly with what was observed in result 1, where indigenous norms and rituals constitute the main support for water management in the community. The

baseline scenario, therefore, represents a hypothetical condition where that support disappears.

In the ethnohydrological scenario, in which cultural management and indigenous norms are activated, farmer pressure grows more slowly than in the baseline scenario, although it continues to follow an upward trend. As a result,

sacred sites decline, but do so less abruptly. This behavior replicates what was found in the first result, indigenous norms, *pagamentos*, *saneamientos*, and access restrictions play an important role in the spiritual and material protection of the territory, but have limited scope in the face of external pressures or rapid environmental changes. In this scenario, the agricultural mosaic increases and the forest decreases, creating a more open and intervened environment. In this more exposed context, indigenous norms manage to slow down the deterioration of sacred sites, but not to reverse or stabilize it. The system remains vulnerable.

Finally, in the co-governance scenario, which simultaneously integrates the three dimensions of the TEK Index with higher levels of cooperation, the trajectories of the system show differentiated behavior. Farmer pressure increases more slowly and never reaches extreme levels. Indigenous norms remain robust, and their interaction with state and community regulation allows the condition of sacred sites to stop falling abruptly and tend toward stable, albeit low, values. This stabilization is consistent with the findings of the TEK Index and socio-environmental transformations, which showed that cultural management has a strong capacity for symbolic regulation but requires external support to sustain itself in the face of increasing pressures. It also responds to observations that agricultural intensification and population growth transform the landscape, making a broader regulatory arrangement necessary to prevent the deterioration of key cultural and ecological elements. The co-governance scenario, therefore, is the only arrangement in which pressure does not skyrocket, Indigenous norms are consolidated in an intercultural framework, and sacred sites achieve a condition of relative persistence.

Taken together, the three scenarios show that, although cultural management and indigenous norms are fundamental to the protection of sacred sites, their capacity to sustain them is reduced when the territory experiences rapid environmental changes and growing demographic pressures. However, when these norms are articulated with institutional and community mechanisms, water management scenarios are configured that can mitigate deterioration and preventing the collapse of the spiritual and ecological system. Figure 5 summarizes this dynamic, showing that co-governance is the arrangement that offers the best conditions for the persistence of sacred sites in a changing landscape.

Discussion of Results

The results of this study confirm that TEK associated with water management in the study area operates as a complex normative and relational system that is currently endangered by socio-environmental and institutional pressures

that limit its capacity to sustain water governance. The low added value of the TEK Index (1.28) does not indicate the disappearance of knowledge, but rather an erosion of its institutional effectiveness, consistent with the literature that suggests that TEK weakens when socio-environmental conditions are altered (Berkes, 2018; Maclean, 2015; Coletta et al., 2024). In this sense, the empirical results are consistent with approaches that conceive of TEK not as a static asset, but as a situated practice whose viability depends on the integrity of the spaces, authorities, and co-governance arrangements that sustain it.

A disaggregated analysis of the index shows that cultural management is the relatively most robust dimension of TEK, confirming the centrality of the Law of Origin and ritual practices in water regulation. This finding is consistent with ethnographic studies documenting how rituality, sacred sites, and spiritual norms function as social institutions that regulate access, reinforce collective responsibilities, and promote conservation (Arsenault, 2021; Datta et al., 2025; Moggridge et al., 2022). However, the intermediate value of this dimension and community perceptions of the loss of sacred sites indicate that cultural strength is insufficient to sustain the system on its own, particularly when intergenerational transmission weakens and external pressures intensify. This is consistent with the literature that points out that the effectiveness of TEK declines when landscape changes and demographic dynamics exceed the scale of community control (Latulippe & McGregor, 2022; Reyes-García et al., 2019).

The weakness of the government norms dimension reinforces this interpretation. The lack of recognition and appropriation of instruments such as POMCA cannot be read solely as an information gap, but rather as an intercultural disconnect between state water planning and indigenous ontologies of water. This result is in line with criticisms of Integrated Water Resources Management (IWRM), which point to its tendency to translate water into a technical-administrative object, rendering invisible cultural, spiritual, and normative dimensions that are central to Indigenous territories (Hoverman & Ayre, 2012; Jackson, 2018; Ngonyani et al., 2024). In contrast, the intermediate performance of indigenous norms shows a coherent internal system, with clear rules and restorative sanctions, but with limited capacity to address external impacts, which coincides with hydro-social approaches that emphasize asymmetries of power and territorial control in water governance (Arsenault, 2021; Moggridge et al., 2022).

The traditional practices identified provide a key insight for the theoretical debate, water management is not expressed as an isolated technical action, but is anchored in specific sacred places and sustained by differentiated authorities. This operational logic reinforces the relevance

of ethnohydrological approaches that emphasize the co-production between biophysical, normative, and cultural dynamics, and the central role of sacred spaces as nodes of territorial regulation (Acharibasam et al. 2024a, 2024b; Diver et al. 2022). The empirical association between the deterioration of these places and the erosion of TEK confirms that knowledge is affected not only by cultural changes, but also by the loss of the material spaces where it is practiced and reproduced.

These findings also resonate with literature on indigenous ontologies and cosmopolitics, which argues that water and sacred sites cannot be reduced to passive environmental resources, but constitute living territorial entities embedded within relational systems of governance, spirituality, and authority (De La Cadena, 2010; Salmond, 2014; Strang, 2004). From this perspective, the limited effectiveness of state planning instruments in the basin is not only an institutional problem, but also reflects tensions between different ways of understanding water, territory, and environmental regulation.

These relational understandings of water also resonate with broader debates on the Rights of Nature, which recognize rivers and ecosystems as living territorial entities with cultural, spiritual, and ecological significance beyond their utilitarian functions.

The dynamics of land use change provide strong empirical support for these tensions. The sustained loss of forest cover, the expansion of the agricultural mosaic, and the increase in infrastructure around the river and sacred sites confirm that the drivers of transformation identified globally operate locally in the basin (IPBES, 2024). The overlap between transformed areas and sacred sites reinforces a critical mechanism: ecological degradation and the erosion of TEK are coupled processes that feed back into each other, simultaneously affecting water regulation and cultural governance of the territory.

Ethnohydrological modeling allows these findings to be integrated. The negative scenario reproduces a trajectory of collapse characterized by accelerated increases in farmer pressure and the almost total loss of sacred sites, which coincides with the socio-hydrological literature on systems without effective regulation (Budiman et al., 2023; Kumar & Saizen, 2023). The ethnohydrological scenario shows that cultural management and indigenous norms act as buffers, slowing down degradation but failing to stabilize the system, reproducing the empirical pattern observed in the TEK Index. In contrast, the co-governance scenario emerges as the only arrangement capable of avoiding collapse trajectories, by articulating indigenous norms, state regulation, and cooperation, in line with approaches of adaptive governance and knowledge integration (Folke et al., 2005; Tengö et al., 2014).

In practical terms, co-governance in the basin would require formal mechanisms of coordination between

Indigenous authorities, environmental agencies, municipal governments, and local water users. This would involve recognizing indigenous authorities as legitimate territorial actors in watershed planning processes, incorporating sacred sites into environmental management instruments, and promoting intercultural dialogue spaces to address conflicts related to agricultural expansion, water use, and territorial regulation.

Conclusions

This study confirms that the TEK of the Arhuaco people plays a central role in water governance in the upper Ari-guaní River basin, based on a relational ontology that conceives of water, the forest, and sacred sites as living and interdependent entities. The cultural practices identified continue to guide decision-making through spiritual and political authorities, supporting the hypothesis that TEK contributes significantly to water management, particularly through the protection of sacred sites as hubs of indigenous water governance.

However, the results show that this contribution is conditional. The TEK Index, together with the analysis of land use change and population dynamics, shows that TEK manages to slow down the impacts associated with deforestation, agricultural expansion, and population growth, but does not fully mitigate them when these pressures intensify. Cultural water management is thus exercised in an ecologically transformed territory, which limits its regulatory capacity when operating in isolation.

Socio-hydrological modeling reinforces this finding by showing that scenarios based exclusively on indigenous norms cushion the deterioration of the system, but do not achieve stable conditions under increasing pressures. In contrast, co-governance scenarios, which articulate TEK with institutional mechanisms, make it possible to reduce pressure and prevent the collapse of sacred sites, demonstrating that the effectiveness of TEK is enhanced when it is inserted into multiscale regulatory arrangements.

The results suggest that water and environmental management instruments should recognize TEK (its practices, authorities, and sacred sites) as functional institutions of water governance and move toward intercultural co-governance schemes that articulate indigenous and state norms without subordination. Likewise, the need to implement territorial strategies differentiated by watershed section is highlighted, prioritizing the protection of forests and waterways in critical areas. Finally, ethnohydrological modeling emerges as a useful tool to support intercultural dialogue and decision-making processes, rather than as a predictive exercise.

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Author Contributions V.G. Conceptualization, Methodology, Field data collection, Data analysis, Writing, Review, and Editing; J.H. Field data collection, Data systematization, and Writing; V.P. Spatial analysis, Dynamic modeling, Data analysis, and Writing; A.G. Review; M.T. Review.

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Data Availability The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Declarations

Ethics This project was submitted to and approved by the Research and Ethics Committee of the Faculty of Architecture and Design at the Pontificia Universidad Javeriana, Colombia (Act No. 151, April 9, 2024). The study was classified as "research without risk" in accordance with national regulations governing research involving human participants. All procedures performed in this study complied with relevant national scientific, technical and ethical standards. Participants ranged in age from 20 to 67 years, and no minors were included in the study. Written informed consent was obtained from all individual participants involved in the study. Authorization to conduct fieldwork was formally granted by the Mamus, as the spiritual authorities, and by the Indigenous cabildos of the Arhuaco territory in the Sierra Nevada de Santa Marta prior to data collection.

Clinical Trial Number Not applicable.

Competing interests The authors declare no competing interests.

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